

Data Quality Assessment Report - Q4 2024

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Approval

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1. EXECUTIVE SUMMARY

1.1 Overall Data Quality Score

In Q4 2024, the Data Quality team conducted comprehensive profiling of 150 Million Records across 58 integrated systems (7 Tier 1, 19 Tier 2, 32 Tier 3). This quarterly assessment represents the most extensive data quality evaluation conducted by Company A to date.

Overall Data Quality Score: 92.4% (Up from 89.0% in Q3 2024)

This score reflects a +3.4 percentage point improvement quarter-over-quarter, demonstrating the effectiveness of targeted remediation efforts and proactive data quality controls.

Overall Score Composition

Component	Weight	Score	Contribution
Completeness	25%	94.2%	23.6%
Accuracy	25%	91.8%	23.0%
Consistency	20%	93.5%	18.7%
Validity	15%	89.4%	13.4%
Uniqueness	10%	95.1%	9.5%
Timeliness	5%	92.0%	4.6%
Overall Weighted Score	100%		92.4%

Scoring Interpretation

Score Range	Rating	Interpretation
95-100%	Excellent	Data is fit for all purposes; minimal issues
90-94%	Good	Data is fit for most purposes; minor issues exist
85-89%	Acceptable	Data is fit for basic purposes; moderate issues re
80-84%	Poor	Data quality issues impact business processes; urg
<80%	Critical	Data is unreliable; immediate executive attention

Company A's Q4 2024 Score: 92.4% (Good) - Data is fit for most purposes with minor issues requiring attention in specific systems.

1.2 Key Highlights & Improvements

Significant Achievements

- 67% Reduction in Critical Issues: Critical data quality issues decreased from 156 (Q4 2023) to 52 (Q4 2024), representing a 67% year-over-year improvement
- Tier 1 System Excellence: All 7 Tier 1 mission-critical systems achieved >90% data quality scores, with Finance (Oracle EBS) reaching 99.1%
- Automated Monitoring Deployed: Informatica Data Quality (IDQ) platform now monitors 1,240 active data quality rules in real-time with automated alerting
- Proactive Issue Detection: 85% of data quality issues were detected proactively through automated profiling (vs 42% in Q1 2024)

■ **Faster Resolution:** Average issue resolution time reduced from 18 days (Q1 2024) to 8 days (Q4 2024), a 56% improvement

Top Performing Systems

System	Domain	Q4 2024 Score	Q3 2024 Score	Improvement
Oracle E-Business Suite	Finance	99.1%	98.5%	+0.6%
SAP SuccessFactors	HR	96.8%	95.2%	+1.6%
Epic EHR	Medical	96.2%	95.8%	+0.4%
Oracle Data Warehouse	Analytics	95.4%	93.1%	+2.3%
Cerner HIS	Medical	94.8%	93.9%	+0.9%

Major Improvement Initiatives Completed in Q4

- "Patient Contact Accuracy Campaign": Contacted 2,400 patients to verify and update mobile numbers; improved EHR contact accuracy from 88% to 94%
- "Vendor Master Data Cleansing": Standardized vendor addresses and deduplicated 180 vendor records; improved Vendor MDM score from 78% to 85%
- "Automated Data Entry Validation": Implemented real-time validation rules in 5 web forms; reduced invalid data entry by 72%
- "Reference Data Standardization": Centralized country codes, city names, and department codes; improved consistency from 87% to 93.5%

1.3 Critical Issues Requiring Attention

Despite overall strong performance, the following critical data quality issues require immediate executive attention:

Priority 1: CRM System Data Quality (79.5% Score - Below Target)

Issue: Customer CRM (Salesforce) scored 79.5%, significantly below the 90% target threshold

Impact:

- 25% of customer email addresses fail validation (e.g., `user@gmail..com`, `test@@domain.com`)
- 18% of customer phone numbers are formatted inconsistently (international format vs local format)
- Unable to execute effective email marketing campaigns (22% bounce rate)

Root Cause:

- "Quick Lead" web form lacks real-time email validation
- No standardized phone number formatting enforced
- Data entry training insufficient for sales team

Remediation:

- Implement real-time email validation API (Target: Jan 15, 2025)
- Deploy automated phone number standardization (Target: Feb 1, 2025)
- Mandatory data entry training for all sales staff (Target: Jan 31, 2025)

Owner: IT Development Lead + Sales Operations Manager

Priority 2: Inactive Contact Information (94% Mobile Number Accuracy)

Issue: 6% of patient mobile numbers in Epic EHR are inactive or disconnected

Impact:

- Cannot deliver critical lab results and appointment reminders via SMS
- Regulatory requirement to notify patients within 24 hours not met for 6% of cases
- Patient satisfaction scores impacted (complaints about missed appointments)

Root Cause:

- Patients do not proactively update contact information
- No automated verification of mobile numbers at check-in
- Mobile numbers collected at first visit (5+ years ago for some patients)

Remediation:

- Implement SMS verification at patient check-in (Target: Feb 15, 2025)
- Launch "Update Your Info" patient portal campaign (Target: Jan 30, 2025)
- Automated SMS bounce detection and flagging (Target: Q1 2025)

Owner: Medical Services Director + IT Operations Manager

Priority 3: Legacy System Data Gaps

Issue: 3 legacy systems (Archive System X, Old Billing, Legacy Pharmacy) have data quality scores <85%

Impact:

- Historical data queries return incomplete results
- FOI (Freedom of Information) requests delayed due to data gaps
- Audit trail incomplete for regulatory compliance

Root Cause:

- Systems scheduled for retirement; no active data quality investment
- Manual data entry without validation
- Lack of integration with master data management (MDM)

Remediation:

- Accelerate system retirement timeline (Target: Q2 2025)
- One-time data cleansing and migration to modern platforms (Target: Q3 2025)
- Interim workaround: Manual data verification for critical queries

Owner: IT Modernization Program Manager

1.4 Year-over-Year Progress

The following table illustrates sustained data quality improvement across all data domains:

YoY Data Quality Score Comparison (Q4 2023 vs Q4 2024)

Data Domain	Systems	Q4 2023	Q4 2024	Change	Status
Finance	Oracle EBS, Hyperion	96.8%	99.1%	+2.3%	■ Excellent
Medical	EHR, HIS, LIS, Pharmacy, PACS	95.5%	96.2%	+4.7%	■ Excellent
HR	SAP SuccessFactors, ADP Workday	94.2%	96.8%	+7.6%	■ Excellent
Operations	Maximo, Fleet, Facilities	84.5%	91.0%	+6.5%	■ Good
Procurement	SAP Ariba, Contracts	88.1%	93.2%	+5.1%	■ Good
Analytics	Data Warehouse, Power BI	86.4%	95.4%	+9.0%	■ Excellent
Customer	Salesforce CRM, Portal	72.8%	79.5%	+6.7%	■ Critical (Below Target)

Analysis: All 7 data domains show positive year-over-year improvement, with the most significant gains in Analytics (+9.0%), HR (+7.6%), and Operations (+6.5%). While Customer CRM improved by 6.7%, it remains below the 90% target threshold and requires focused remediation in Q1 2025.

Critical Issue Volume Trends

Quarter	Critical Issues	High Issues	Medium Issues	Low Issues	Total Issues
Q4 2023	156	342	1,204	2,890	4,592
Q1 2024	124	298	1,156	2,650	4,228
Q2 2024	89	267	1,025	2,420	3,801
Q3 2024	68	234	945	2,180	3,427
Q4 2024	52	198	862	1,950	3,062
YoY Change	-67%	-42%	-28%	-33%	-33%

Key Observation: The organization has achieved a 67% reduction in critical data quality issues year-over-year, demonstrating the effectiveness of proactive data quality management and systematic remediation efforts.

2. ASSESSMENT METHODOLOGY

2.1 Scope of Assessment

2.1.1 Systems Covered

The Q4 2024 assessment covered 58 integrated systems across all business domains:

Tier 1 Systems (Mission-Critical): 7 systems

- Electronic Health Records (EHR) - Epic Systems
- Hospital Information System (HIS) - Cerner Millennium
- Laboratory Information System (LIS) - Sunquest
- Financial Management System - Oracle E-Business Suite R12
- Human Resources Management System - SAP SuccessFactors
- Data Warehouse - Oracle Exadata
- Business Intelligence Platform - Microsoft Power BI

Tier 2 Systems (Business-Critical): 19 systems

- Customer CRM (Salesforce), Procurement (SAP Ariba), Pharmacy Management (McKesson), Radiology PACS (GE Healthcare), Document Management (SharePoint), and 14 additional systems

Tier 3 Systems (Departmental): 32 systems

- Legacy and departmental applications including Archive System X, Old Billing System, Legacy Pharmacy, and 29 additional specialized systems

2.1.2 Data Volume Profiled

Data Category	Record Count	File/Table Count	Volume (GB)
Patient Medical Records	18,500 active patients	142 tables	85 GB
Laboratory Results	820,000 test results	28 tables	42 GB
Financial Transactions	2,800,000 transactions	68 tables	124 GB
Employee HR Records	850 employees	35 tables	8 GB
Vendor Master Data	4,200 vendors	12 tables	6 GB
Customer CRM Records	95,000 customers	48 tables	32 GB
Operational Data	3,500,000 records	185 tables	220 GB
Analytics/BI Data	142,000,000 rows	380 tables	1,850 GB
TOTAL	~150 Million Records	898 tables/files	2,367 GB

2.1.3 Assessment Period

- Profiling Window: October 1 - November 30, 2024 (8 weeks)
- Analysis & Reporting: December 1-28, 2024 (4 weeks)
- Total Assessment Duration: 12 weeks

2.2 Data Quality Dimensions

The assessment evaluated data quality across six standard dimensions aligned with industry best practices (DAMA-DMBOK framework):

1. Completeness

Definition: The extent to which required data is present (no missing values)

Measurement: % of records with all mandatory fields populated

Target Threshold: $\geq 95\%$ for Critical Data Elements (CDEs)

Example Rules:

- Patient records must have National ID
- Financial transactions must have Vendor ID
- Employee records must have Department Code

2. Accuracy

Definition: The extent to which data correctly represents the real-world entity

Measurement: % of records that match authoritative sources or pass verification checks

Target Threshold: $\geq 95\%$ for CDEs

Example Rules:

- Mobile numbers must be verified (SMS bounce detection)
- Email addresses must be deliverable (SMTP verification)
- Addresses must be geocodable (Google Maps API validation)

3. Consistency

Definition: The extent to which data is consistent across systems and over time

Measurement: % of records that have matching values across related systems

Target Threshold: $\geq 95\%$ for master data

Example Rules:

- Vendor Name in Oracle EBS must match Vendor Name in SAP Ariba
- Customer ID in CRM must match Customer ID in Billing
- Employee Department in HR must match Department in Payroll

4. Validity

Definition: The extent to which data conforms to defined formats, ranges, and business rules

Measurement: % of records that pass validation rules (regex, range checks, reference data)

Target Threshold: $\geq 95\%$ for all fields with validation rules

Example Rules:

- Email addresses must match regex pattern `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`
- Dates must be in range (e.g., Birth Date must be between 1920-2024)
- Country codes must exist in reference data table (ISO 3166-1 alpha-2)

5. Uniqueness

Definition: The extent to which records are unique (no duplicates)

Measurement: % of records that are not duplicated based on business key

Target Threshold: 100% for entities with unique identifiers (e.g., invoices, patients, employees)

Example Rules:

- Invoice Number must be unique in Financial System
- National ID must be unique in Patient Records
- Employee ID must be unique in HR System

6. Timeliness

Definition: The extent to which data is current and up-to-date

Measurement: % of records that have been updated within expected timeframe

Target Threshold: $\geq 90\%$ for time-sensitive data

Example Rules:

- Patient contact information must be updated within 12 months
- Vendor bank account details must be verified annually
- Employee job title must reflect current role (updated within 30 days of change)

2.3 Profiling Approach

Automated Profiling (Primary Method)

Technology Platform: Informatica Data Quality (IDQ) v10.5

Profiling Frequency:

- Critical Data Elements (CDEs): Real-time monitoring (continuous)

- High-Priority Data: Daily automated profiling
- Standard Data: Weekly automated profiling

Profiling Outputs:

- Column-level statistics (min, max, mean, median, mode, standard deviation)
- Pattern analysis (regex patterns, frequency distributions)
- Null value analysis (% null, % blank, % default values)
- Uniqueness analysis (distinct count, duplicate count)
- Referential integrity checks (foreign key validation)

Manual Validation (Secondary Method)

Sampling Approach: Statistical sampling used for systems with high transaction volumes

Sample Size Calculation:

- Confidence Level: 95%
- Margin of Error: $\pm 3\%$
- Minimum Sample Size: 1,068 records per system (for populations >100,000)

Manual Validation Scenarios:

- Visual inspection of data patterns
- Business rule verification (complex logic not automatable)
- Cross-system consistency checks (manual matching)

2.4 Scoring Methodology

Dimension-Level Scoring

For each data quality dimension (Completeness, Accuracy, Consistency, Validity, Uniqueness, Timeliness):

Dimension Score = (Number of Passing Records / Total Records) $\times$ 100%

Example:

- Total Patient Records: 18,500
- Records with National ID populated: 18,130
- Completeness Score = (18,130 / 18,500) $\times$ 100% = 98.0%

System-Level Scoring

System-level scores are calculated as a weighted average of dimension scores:

System Score = Σ (Dimension Score $\times$ Dimension Weight)

Where:

- Completeness Weight: 25%
- Accuracy Weight: 25%

- Consistency Weight: 20%
- Validity Weight: 15%
- Uniqueness Weight: 10%
- Timeliness Weight: 5%

Domain-Level Scoring

Domain-level scores (e.g., Finance, Medical, HR) are calculated as:

Domain Score = Σ (System Score × System Record Count) / Total Domain Record Count

This weighted average ensures that systems with higher data volumes have proportionally greater influence on the domain score.

Overall Organizational Score

Overall Score = Σ (Domain Score × Domain Weight)

Domain weights are determined by business criticality:

- Medical: 30%
- Finance: 25%
- HR: 15%
- Operations: 10%
- Procurement: 10%
- Customer: 10%

3. SCORECARD SUMMARY

3.1 Domain-Level Scorecard

Domain	Systems	Record Count	Q4 2024 Score	Q3 2024 Score	Change	Status
Finance	Oracle EBS, Hyperion Financials	2,800,000	99.1%	98.5%	+0.6%	■ Excellent
Medical	EHR, HIS, LIS, Pharmacy	19,320,000	96.2%	95.8%	+0.4%	■ Excellent
HR	SAP SuccessFactors, ADP Payroll	450,000	96.8%	95.2%	+1.6%	■ Excellent
Analytics	Data Warehouse, Power BI	12,000,000	95.4%	93.1%	+2.3%	■ Excellent
Procurement	SAP Ariba, Contracts	7,800	93.2%	91.5%	+1.7%	■ Good
Operations	Maximo, Fleet, Facilities	3,500,000	91.0%	89.4%	+1.6%	■ Good
Customer	Salesforce CRM, Patient Portal	15,000,000	79.5%	76.2%	+3.3%	■ Critical

3.2 System-Level Scorecard (Top 15 Systems)

Rank	System	Domain	Record Count	Q4 2024 Score	Q3 2024 Score	Change	Status
1	Oracle E-Business Suite	Finance	2,800,000	99.1%	98.5%	+0.6%	■
2	SAP SuccessFactors	HR	850	96.8%	95.2%	+1.6%	■
3	Epic EHR	Medical	18,500	96.2%	95.8%	+0.4%	■
4	Oracle Data Warehouse	Analytics	142,000,000	95.4%	93.1%	+2.3%	■
5	Cerner HIS	Medical	5,200,000	94.8%	93.9%	+0.9%	■
6	Sunquest LIS	Medical	820,000	94.2%	92.8%	+1.4%	■
7	SAP Ariba	Procurement	4,200	93.2%	91.5%	+1.7%	■
8	McKesson Pharmacy	Medical	1,450,000	92.8%	91.2%	+1.6%	■
9	GE PACS (Radiology)	Medical	1,200,000	92.5%	91.8%	+0.7%	■
10	IBM Maximo (Assets)	Operations	3,200,000	91.0%	89.4%	+1.6%	■
11	Microsoft Power BI	Analytics	180 reports	90.2%	88.5%	+1.7%	■
12	SharePoint	Operations	285,000	89.5%	87.9%	+1.6%	■
13	ADP Payroll	HR	850	88.2%	86.5%	+1.7%	■
14	Archive System X	Operations	58,000	82.4%	80.1%	+2.3%	■
15	Salesforce CRM	Customer	95,000	79.5%	76.2%	+3.3%	■

- Legend:
- ■ Good (≥90%)
 - ■ Acceptable (85-89%)
 - ■ Critical (<85%)

3.3 Dimension-Level Scorecard

Dimension	Weight	Q4 2024 Score	Q3 2024 Score	Change	Status
Uniqueness	10%	95.1%	94.8%	+0.3%	■ Excellent
Completeness	25%	94.2%	92.5%	+1.7%	■ Excellent
Consistency	20%	93.5%	91.2%	+2.3%	■ Good
Timeliness	5%	92.0%	90.8%	+1.2%	■ Good
Accuracy	25%	91.8%	89.6%	+2.2%	■ Good
Validity	15%	89.4%	86.5%	+2.9%	■ Acceptable

Weighted Overall Score: 92.4% (Up from 89.0% in Q3 2024)

4. DETAILED FINDINGS (BY SYSTEM)

4.1 Tier 1: Electronic Health Records (EHR) - Epic

System Owner: Dr. Yousef Al-Harbi (Medical Services Director)

Data Steward: Dr. Maha Al-Zahrani

Overall Score: 96.2% (Excellent)

Volume & Scope

Metric	Value
Total Patient Records	18,500 active patients
Total Encounters	284,000 (2024 YTD)
Data Tables Profiled	142 tables
Data Volume	85 GB
Critical Data Elements (CDEs)	48 fields

Dimension Scores

Dimension	Score	Status	Key Findings
Completeness	98.0%	■	National ID missing for 2% (370 patients) - mostly
Accuracy	94.0%	■	6% (1,110 patients) have inactive mobile numbers (
Consistency	97.5%	■	Patient demographics consistent with national iden
Validity	95.8%	■	Date ranges valid; minor issues with future appoin
Uniqueness	99.5%	■	92 potential duplicate patient records (0.5%) flag
Timeliness	96.2%	■	Contact information updated for 96.2% of patients

Critical Issues

Issue #EHR-001: Inactive Mobile Numbers (High Priority)

- Description: 6% of patient mobile numbers (1,110 records) are inactive or disconnected based on SMS bounce detection
- Business Impact:
 - Cannot deliver critical lab results via SMS (regulatory requirement to notify within 24 hours)
 - Appointment reminder SMS failed for 1,110 patients in Q4 (contributed to 8% no-show rate)
 - Patient satisfaction scores impacted (15 complaints logged in Q4 about missed notifications)

- Root Cause:
- Patients do not proactively update contact information when changing mobile numbers
- No automated verification at patient check-in (verification manual and inconsistently performed)
- Some patient records 5+ years old with original mobile numbers
- Remediation Plan:
- Action 1: Implement automated SMS verification at patient check-in (validate via OTP code) - Target: Feb 15, 2025
- Action 2: Launch "Update Your Contact Info" campaign via patient portal and email - Target: Jan 30, 2025
- Action 3: Automated SMS bounce detection and flagging in EHR with staff alert - Target: Q1 2025
- Owner: Dr. Yousef Al-Harbi (Medical Services) + IT Operations Manager

Issue #EHR-002: Missing National IDs (Medium Priority)

- Description: 2% of patient records (370 patients) lack National ID, primarily emergency admissions
- Business Impact:
- Unable to query national health insurance database for coverage verification
- Cannot integrate with national immunization registry
- Billing delays for government-funded healthcare programs
- Root Cause:
- Emergency admissions prioritize immediate care over administrative data collection
- National ID not mandatory field in EHR workflow (can be bypassed)
- Follow-up data collection process exists but not consistently executed
- Remediation Plan:
- Action 1: Make National ID mandatory field for all patient registrations (emergency bypass allowed with required follow-up task) - Target: Q1 2025
- Action 2: Retrospective outreach campaign to collect missing National IDs (call 370 patients) - Target: Feb 28, 2025
- Action 3: Monthly reporting to Medical Services Director on incomplete patient records - Target: Ongoing
- Owner: Patient Registration Manager

Positive Highlights

- Medication Allergy Documentation: 99.8% of patients have allergy status documented (significant improvement from 92% in 2023)
- Immunization Record Completeness: 97.5% of pediatric patients have complete immunization records
- Clinical Note Quality: 96% of clinical encounters have structured clinical notes (using EHR templates)

4.2 Tier 1: Hospital Information System (HIS) - Cerner

System Owner: Reem Al-Qahtani (Hospital Operations Manager)

Overall Score: 94.8% (Excellent)

Volume & Scope

Metric	Value
Total Inpatient Admissions	24,800 (2024 YTD)
Total Outpatient Visits	185,000 (2024 YTD)
Operating Room Procedures	8,200 (2024 YTD)
Data Tables Profiled	95 tables
Data Volume	62 GB

Dimension Scores

Dimension	Score	Status	Key Findings
Completeness	96.5%	■	Discharge summaries complete for 96.5% of inpatient
Accuracy	93.8%	■	Bed assignment data 93.8% accurate (cross-validated)
Consistency	94.2%	■	Patient demographics consistent with EHR (94.2% exact)
Validity	92.5%	■	Procedure codes valid per ICD-10 standards (92.5% valid)
Uniqueness	98.2%	■	Minimal duplicate admission records (1.8%)
Timeliness	94.0%	■	Discharge summaries completed within 48 hours for 94.0%

Critical Issues

- Issue #HIS-001: Incomplete Discharge Summaries (Medium Priority)
- Description: 3.5% of inpatient discharge records (867 discharges) lack complete discharge summaries
 - Business Impact:
 - Delays in billing submission (requires complete clinical documentation)
 - Compromises continuity of care for follow-up visits
 - Audit compliance risk (regulatory requirement for discharge summary within 48 hours)
 - Root Cause:
 - Physician workflow allows discharge summary bypass (completed retrospectively)
 - Busy physicians prioritize patient care over administrative documentation
 - No automated reminder system for pending discharge summaries
 - Remediation Plan:
 - Action 1: Implement automated reminder system in EHR (daily notifications to physicians) - Target: Q1 2025
 - Action 2: Enforce discharge summary completion before final billing - Target: Q2 2025
 - Action 3: Monthly physician performance dashboard showing discharge summary compliance - Target: Ongoing
 - Owner: Chief Medical Officer

Positive Highlights

- Operating Room Data Quality: 98.2% data quality for OR scheduling and procedure documentation

■ Medication Administration Records: 97.8% of administered medications documented with timestamp and nurse signature

4.3 Tier 1: Laboratory Information System (LIS) - Sunquest

System Owner: Layla Al-Fahad (Laboratory Services Manager)
Overall Score: 94.2% (Excellent)

Volume & Scope

Metric	Value
Total Lab Tests (2024)	820,000 tests
Test Types	245 different test types
Data Tables Profiled	28 tables
Data Volume	42 GB

Dimension Scores

Dimension	Score	Status	Key Findings
Completeness	95.8%	■	Test results complete; minor gaps in specimen collection
Accuracy	93.2%	■	6.8% of results flagged for quality control review
Consistency	94.5%	■	Reference ranges consistent with clinical guidelines
Validity	92.8%	■	Test codes valid per LOINC standard (92.8% compliance)
Uniqueness	99.0%	■	Minimal duplicate test orders (1.0%)
Timeliness	93.5%	■	Results delivered within TAT (Turnaround Time) for 93.5%

Positive Highlights

- Critical Result Notification: 99.5% of critical lab results communicated to ordering physician within 1 hour
- Quality Control Documentation: 98.8% of quality control tests documented and within acceptable ranges

4.4 Tier 1: Financial Management - Oracle E-Business Suite

System Owner: Fahad Al-Shammari (Finance Director)
Data Steward: Fahad Al-Qahtani
Overall Score: 99.1% (Excellent - Highest Score)

Volume & Scope

Metric	Value
Total Transactions (2024)	2,800,000
Invoices Processed	184,000
Purchase Orders	28,500
Data Tables Profiled	68 tables
Data Volume	124 GB

Dimension Scores

Dimension	Score	Status	Key Findings
Completeness	99.8%	■	All mandatory financial fields populated; exceptio
Accuracy	99.2%	■	Account coding accuracy 99.2% (validated through m
Consistency	98.5%	■	Vendor names 98.5% consistent with Procurement sy
Validity	99.0%	■	GL account codes 99% valid per chart of accounts
Uniqueness	100.0%	■	Zero duplicate invoices detected (invoice number u
Timeliness	98.8%	■	Invoices processed within 10 days for 98.8% of sub

Critical Issues

No Critical Issues Identified - Oracle E-Business Suite maintains exceptional data quality standards

Positive Highlights

- Perfect Uniqueness: Zero duplicate invoices or purchase orders detected in Q4 2024
- Automated Validation: Strong preventive controls at data entry (mandatory fields, dropdown validations, referential integrity)
- Monthly Reconciliation: Systematic monthly reconciliation ensures ongoing data accuracy

4.5 Tier 1: Human Resources - SAP SuccessFactors

System Owner: Nora Al-Mutairi (HR Director)
Data Steward: Nora Al-Harbi
Overall Score: 96.8% (Excellent)

Volume & Scope

Metric	Value
Total Employees	850 active employees
Total HR Transactions (2024)	12,400 (promotions, transfers, leaves, etc.)
Data Tables Profiled	35 tables
Data Volume	8 GB

Dimension Scores

Dimension	Score	Status	Key Findings
Completeness	98.5%	■	Employee personal information 98.5% complete
Accuracy	96.2%	■	Employee contact information verified annually (96%)
Consistency	97.0%	■	Employee department codes consistent with organizational chart
Validity	95.8%	■	Job titles conform to standard job catalog (95.8%)
Uniqueness	100.0%	■	Zero duplicate employee records (National ID unique)
Timeliness	95.5%	■	Organizational changes reflected in HR system within 2 weeks

Positive Highlights

- Perfect Uniqueness: Zero duplicate employee records (National ID enforced as unique identifier)
- Comprehensive Training Records: 96% of employees have complete training history documented

4.6 Tier 1: Data Warehouse - Oracle Exadata

System Owner: Khalid Al-Harbi (Data Warehouse Manager)

Overall Score: 95.4% (Excellent)

Volume & Scope

Metric	Value
Total Records	142,000,000 rows
Source Systems Integrated	58 source systems
Data Tables	380 tables (fact and dimension tables)
Data Volume	1,850 GB

Dimension Scores

Dimension	Score	Status	Key Findings
Completeness	96.8%	■	ETL processes maintain high completeness; minor gaps in legacy data
Accuracy	94.5%	■	Data transformations validated; 94.5% accuracy maintained
Consistency	96.2%	■	Cross-system consistency enforced through MDM integration
Validity	93.8%	■	Business rules enforced during ETL; 93.8% validity score
Uniqueness	95.0%	■	Deduplication rules applied; 5% duplicates from source
Timeliness	94.2%	■	Data refreshed daily; 94.2% of data <24 hours old

Positive Highlights

- Comprehensive Data Lineage: 90% of data flows documented with automated lineage in Informatica EDC
- Systematic Data Quality Checks: 1,240 data quality rules applied during ETL processes

4.7 Tier 2: Customer CRM - Salesforce

System Owner: Abdullah Al-Rashid (Marketing & Engagement Director)
Overall Score: 79.5% (Critical - Below Target)

Volume & Scope

Metric	Value
Total Customer Records	95,000 customers
Total Opportunities	8,500 (2024 YTD)
Data Tables Profiled	48 tables
Data Volume	32 GB

Dimension Scores

Dimension	Score	Status	Key Findings
Completeness	82.5%	■	17.5% of records have incomplete contact information
Accuracy	75.2%	■	Email bounce rate 22%, phone number accuracy 78%
Consistency	80.5%	■	Customer names inconsistent formatting (variations)
Validity	74.8%	■	25% of emails fail regex validation (e.g., user@gm)
Uniqueness	88.0%	■	12% potential duplicate customer records
Timeliness	76.2%	■	24% of records not updated in past 12 months (stale)

Critical Issues

Issue #CRM-001: Invalid Email Addresses (CRITICAL PRIORITY)

- Description: 25% of customer email addresses (23,750 records) fail validation checks
- Examples: `user@gmail..com`, `test@@domain.com`, `email@domain`, `user@.com`
- Business Impact:
 - Email marketing campaigns have 22% bounce rate (industry average: 2-5%)
 - Unable to deliver appointment confirmations and health education materials via email
 - Wasted marketing budget (sending to invalid addresses)
 - Brand reputation risk (high bounce rates damage sender reputation scores)
- Root Cause:
 - "Quick Lead" web form on public website lacks real-time email validation
 - Manual data entry by sales team without validation checks
 - Data imported from legacy systems without cleansing
- Remediation Plan:
 - Action 1 (URGENT): Implement real-time email validation API on "Quick Lead" web form - Target: Jan 15, 2025
 - Action 2: Deploy email verification service (ZeroBounce or similar) to validate existing database - Target: Feb 1, 2025
 - Action 3: Mandatory data entry training for all sales staff (email format standards) - Target: Jan 31, 2025
 - Action 4: Monthly email deliverability reporting to Sales Director - Target: Ongoing
- Owner: IT Development Lead + Sales Operations Manager
- Budget: SAR 45,000 (ZeroBounce API subscription + validation service)

Issue #CRM-002: Inconsistent Phone Number Formatting (HIGH PRIORITY)

- Description: Phone numbers stored in multiple formats (international, local, no country code)
- Examples: `+966501234567`, `0501234567`, `501234567`, `(050) 123-4567`
- Business Impact:
 - SMS marketing campaigns fail due to formatting issues (18% failure rate)
 - Call center dialer integration requires manual reformatting
 - Unable to perform phone number deduplication (same number in different formats)
- Root Cause:
 - No enforced phone number format standard in Salesforce
 - Multiple data entry points (web form, manual entry, data import) with different formats
- Remediation Plan:
 - Action 1: Implement automated phone number standardization (convert all to +966XXXXXXXXXX format) - Target: Feb 1, 2025
 - Action 2: Enforce phone number format validation on all web forms and manual entry - Target: Feb 15, 2025
 - Action 3: One-time data cleansing of existing 95,000 records - Target: Q1 2025
- Owner: IT Development Lead + CRM Administrator
- Budget: SAR 25,000 (phone number validation API)

Issue #CRM-003: Duplicate Customer Records (MEDIUM PRIORITY)

- Description: 12% potential duplicate customer records (11,400 suspected duplicates)

- Examples: Same person with slight name variations (`Ahmed Ali`, `Ahmed A.`, `A. Ali`)
- Business Impact:
 - Marketing campaigns sent multiple times to same customer (customer complaints)
 - Inaccurate customer segmentation and analytics
 - Sales team confusion (which record is the "master"?)
- Root Cause:
 - No automated duplicate detection rules in Salesforce
 - Sales reps create new records instead of searching for existing ones
 - Data imported from multiple sources without deduplication
- Remediation Plan:
 - Action 1: Implement automated duplicate detection rules in Salesforce (fuzzy matching on name + phone) - Target: Q1 2025
 - Action 2: Conduct one-time deduplication project (manual review + merge of 11,400 records) - Target: Q2 2025
 - Action 3: Mandatory Salesforce training for all sales reps (search before creating new record) - Target: Q1 2025
 - Owner: CRM Administrator + Sales Operations Manager

Positive Highlights

- Opportunity Data Quality: Sales opportunity data (pipeline) maintains 92% data quality (significantly better than customer master data)
- Improvement Trend: CRM data quality improved from 76.2% (Q3) to 79.5% (Q4), demonstrating effectiveness of ongoing remediation efforts

4.8 Tier 2: Procurement - SAP Ariba

System Owner: Omar Al-Mutairi (Procurement Director)
Overall Score: 93.2% (Good)

Volume & Scope

Metric	Value
Total Vendors	4,200 vendors
Active Contracts	3,800 contracts
Purchase Requisitions (2024)	18,500
Data Tables Profiled	24 tables
Data Volume	18 GB

Dimension Scores

Dimension	Score	Status	Key Findings
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Completeness	94.8%	■	Vendor master data 94.8% complete (bank details, t
Accuracy	92.5%	■	Vendor addresses geocodable (92.5% success rate)
Consistency	91.8%	■	Vendor names 91.8% consistent with Finance system
Validity	93.0%	■	Contract amounts within approved budget ranges (93
Uniqueness	96.5%	■	Minimal duplicate vendor records (3.5%)
Timeliness	92.0%	■	Vendor data updated annually (92% compliance)

Positive Highlights

- Contract Compliance: 98.5% of purchase orders comply with contract terms (price, quantity, specifications)
- Vendor Certification: 95% of vendors have required certifications documented (business license, tax registration)

5. SYSTEMATIC ISSUES

5.1 "The Trailing Space Problem"

Issue Description

During Q4 profiling, the Data Quality team discovered 50,000 records across multiple systems where text fields contain trailing spaces or multiple internal spaces.

Examples:

- "Ahmed Ali " (trailing space)
- "Fatima Al-Zahrani" (double space between first and last name)
- " Khalid Al-Rashid" (leading space)

Systems Affected

System	Affected Records	% of Total
Salesforce CRM	18,500	19.5%
Archive System X (Legacy)	12,400	21.4%
SharePoint	8,200	2.9%
Old Billing System (Legacy)	6,800	38.5%
Other Systems	4,100	<5%
TOTAL	50,000	-

Business Impact

- Search Failures: Users searching for "Ahmed Ali" cannot find "Ahmed Ali " (with trailing space)
- Duplicate Detection Failures: Duplicate detection algorithms fail to match `"Ahmed Ali"` and `"Ahmed Ali "`
- Reporting Inaccuracies: GROUP BY queries count `"Ahmed Ali"` and `"Ahmed Ali "` as two different entities
- Integration Failures: Cross-system matching fails when one system has trailing spaces and another doesn't

Root Cause Analysis

1. Legacy Data Migration: Data migrated from old systems contained unclean text fields
2. Manual Data Entry: Users accidentally add trailing spaces when typing (difficult to see visually)
3. Copy-Paste Operations: Users copy data from Excel or Word documents that contain extra spaces
4. No Preventive Controls: Application UI does not automatically trim spaces on data entry

Remediation Plan

Action 1: Implement Global "Trim" Rule in ETL Pipelines

- Description: Add automated TRIM() function to all ETL jobs loading data into Data Warehouse
- Impact: Prevents trailing space problem in centralized analytics environment
- Owner: Data Engineering Team
- Status: ■ COMPLETED (Nov 2024)

Action 2: Retrospective Data Cleansing

- Description: Execute one-time SQL UPDATE statements to trim all text fields in source systems
- Scope: 50,000 records across 5 systems
- Owner: Database Administrators
- Target Date: Q1 2025
- Status: ■ IN PROGRESS (30% complete as of Dec 2024)

Action 3: Implement Preventive Controls in UI

- Description: Modify web forms and application UI to automatically trim user input
- Technology: JavaScript trim() function on form submission
- Owner: Application Development Team
- Target Date: Q1 2025
- Status: ■ PLANNED

Estimated Completion: March 31, 2025

5.2 "Default Date Syndrome"

Issue Description

The Data Quality team discovered 15,000 patient records with Date of Birth (DOB) = 1900-01-01, indicating that users are selecting the default date value rather than entering the actual birth date.

Systems Affected

System	Affected Records	% of Total Patients
Epic EHR	12,200	66%
Patient Portal (Web)	2,800	2.9% (of 95,000 portal users)
TOTAL	15,000	-

Business Impact

- Inaccurate Age Calculations: Reports show patients as 124 years old (2024 - 1900)
- Clinical Decision Support Failures: Age-based clinical alerts fail (e.g., pediatric medication dosing alerts)
- Regulatory Reporting Errors: Age demographics reported to government agencies are inaccurate
- Insurance Claims Rejections: Insurance claims rejected due to impossible patient ages

Root Cause Analysis

1. Default Date in UI: Date picker fields default to 1900-01-01 if user doesn't select a date
2. User Behavior: Users select the default date (first date in picker) to quickly bypass required field validation
3. Training Gap: Staff not trained on importance of accurate DOB for clinical safety
4. Emergency Scenarios: In emergency admissions, staff prioritize immediate care over accurate data entry

Remediation Plan

Action 1: Remove Default Date from UI

- Description: Modify date picker fields to have NO default value (blank until user selects)
- Impact: Forces users to actively select a valid birth date
- Owner: Application Development Team (Epic customization)
- Target Date: Q1 2025
- Status: ■ IN PROGRESS

Action 2: Implement Date Validation Rules

- Description: Add validation rule: DOB must be between 1920-01-01 and Current Date minus 1 day
- Impact: Rejects obviously invalid dates (1900-01-01 will be rejected)
- Owner: Application Development Team
- Target Date: Q1 2025
- Status: ■ IN PROGRESS

Action 3: Retrospective Data Collection

- Description: Contact 15,000 patients to collect actual DOB and update records
- Approach:
 - Phase 1: Call patients scheduled for upcoming appointments (8,000 patients) - Q1 2025
 - Phase 2: Email remaining patients requesting DOB update via patient portal (7,000 patients) - Q2 2025
- Owner: Call Center Manager + Patient Services
- Target Date: Q2 2025

- Status: ■ PLANNED

Estimated Completion: June 30, 2025

5.3 "Inactive Contact Information"

Issue Description

Cross-system validation revealed that a significant percentage of contact information (email addresses, phone numbers) in various systems is inactive or unreachable.

Validation Results

Contact Type	System	Total Records	Inactive	% Inactive
Mobile Numbers	Epic EHR	18,500	1,110	6.0%
Mobile Numbers	Salesforce CRM	95,000	9,500	10.0%
Email Addresses	Salesforce CRM	95,000	20,900	22.0%
Email Addresses	Patient Portal	95,000	8,550	9.0%

Validation Methodology

- Mobile Numbers: SMS bounce detection (automated test SMS sent, bounce rate tracked)
- Email Addresses: SMTP validation (checks if email server accepts messages to the address)

Business Impact

- Communication Failures: Critical notifications (lab results, appointment reminders) fail to deliver
- Marketing Waste: Email marketing budget wasted on undeliverable addresses (22% bounce rate)
- Regulatory Non-Compliance: Healthcare regulations require notification of critical results within 24 hours (failures logged)
- Patient Safety Risk: Patients do not receive critical health information

Remediation Plan

Action 1: Implement Automated Validation

- Mobile Numbers: SMS verification at patient check-in (OTP code sent and verified)
- Email Addresses: Real-time SMTP validation on web forms and manual entry
- Owner: IT Development Team
- Target Date: Q1 2025
- Status: ■ IN PROGRESS

Action 2: Periodic Re-Validation

- Mobile Numbers: Send quarterly test SMS to all patients; flag bounces

- Email Addresses: Send quarterly email campaigns; flag bounces
- Owner: Marketing Operations + IT Operations
- Target Date: Q1 2025 (ongoing quarterly)
- Status: ■ PLANNED

Estimated Completion: March 31, 2025 (initial implementation); ongoing quarterly validation

5.4 "Non-Standard Reference Data"

Issue Description

Multiple systems use different reference data values for the same concepts, causing integration and consistency issues.

Examples

Country Codes:

- Oracle EBS uses: `SA`, `AE`, `KW` (ISO 3166-1 alpha-2)
- Legacy Billing uses: `Saudi Arabia`, `United Arab Emirates`, `Kuwait` (full country names)
- SAP Ariba uses: `SAU`, `ARE`, `KWT` (ISO 3166-1 alpha-3)

Department Codes:

- SAP SuccessFactors (HR): `MED-001`, `FIN-002`, `IT-003`
- Oracle EBS (Finance): `MEDICAL`, `FINANCE`, `IT`
- Salesforce CRM: Free-text field (no standardization)

Systems Affected

- Legacy Procurement System
- Old Billing System
- 8 departmental applications

Business Impact

- Integration Failures: ETL jobs fail when mapping between different reference data formats
- Manual Mapping Required: Data engineers maintain manual mapping tables (time-consuming and error-prone)
- Inconsistent Reporting: Reports aggregate data differently based on which reference data is used
- Master Data Management Challenges: Cannot achieve unified master data with inconsistent reference data

Remediation Plan

Action 1: Establish Central Reference Data Repository

- Description: Create central reference data tables in Data Warehouse for: Countries, Cities, Departments, Job Titles, Product Categories
- Owner: Data Architecture Team

- Target Date: Q1 2025
- Status: ■ COMPLETED

Action 2: Enforce Reference Data in New Projects

- Description: Architecture review gate requires all new projects to use central reference data (via API or database view)
- Owner: Chief Data Officer
- Target Date: Q1 2025
- Status: ■ IN PROGRESS

Action 3: Legacy System Remediation

- Option A: Retire legacy systems (target Q4 2025)
- Option B: Implement API integration to central reference data (if retirement delayed)
- Owner: IT Modernization Program Manager
- Target Date: Q2 2025
- Status: ■ PLANNED

Estimated Completion: Q2 2025 (interim solution); Q4 2025 (permanent solution via system retirement)

5.5 "Duplicate Records Across Systems"

Issue Description

Cross-system analysis revealed that the same real-world entities (customers, vendors, patients) exist as duplicate records across multiple systems with slight variations.

Example: Same Customer in Multiple Systems

System	Customer Name	Customer ID	Email
Salesforce CRM	Ahmed Ali Al-Rashid	CRM-12345	ahmed.ali@email.com
Patient Portal	Ahmed A. Al-Rashid	PAT-98765	ahmed.rashid@email.com
Billing System	Ahmed Ali Rashid	BILL-54321	ahmed@email.com

Analysis: These are likely the same person, but systems treat them as 3 different entities.

Duplicate Detection Results

Entity Type	Systems Analyzed	Suspected Duplicates	% of Total
Customers	CRM, Portal, Billing	8,500	9%
Vendors	Oracle EBS, SAP Ariba	420	10%
Employees	SAP SuccessFactors, Payroll	0	0% (enforced uniqueness)

Business Impact

- 360-Degree View Impossible: Cannot create unified customer view due to duplicate records across systems
- Redundant Communications: Customers receive multiple communications (email, SMS) because they exist as separate records
- Inaccurate Analytics: Customer count and segmentation analytics are inflated due to duplicates
- Master Data Management Challenges: MDM implementation requires extensive deduplication effort

Remediation Plan

Action 1: Implement Customer MDM

- Description: Extend Informatica MDM to Customer domain (currently only Customer MDM active)
- Capability: Central Golden Record for customers with automated matching and merging
- Owner: MDM Manager
- Target Date: Q3 2025
- Status: ■ PLANNED (project approved, vendor engaged)

Action 2: Vendor MDM Data Cleansing

- Description: Deduplicate 420 suspected vendor duplicates through manual review + automated matching
- Owner: Procurement Manager + MDM Team
- Target Date: Q1 2025
- Status: ■ IN PROGRESS (40% complete)

Estimated Completion: Q3 2025 (Customer MDM); Q1 2025 (Vendor deduplication)

6. ROOT CAUSE ANALYSIS

6.1 Process-Related Root Causes

RC-PROC-001: Insufficient Data Entry Training

Description: Staff receive minimal training on data entry best practices and the importance of data quality

Evidence:

- Survey of 120 data entry staff: 68% report <2 hours of data entry training
- No standardized training curriculum for data entry
- Training completion rate: 72% (below 85% target)

Impact Systems: Salesforce CRM, Patient Portal, Legacy Systems

Remediation:

- Develop comprehensive data entry training curriculum (Target: Q1 2025)
- Mandatory training for all staff with data entry responsibilities (Target: Q1 2025)

- Annual refresher training (Target: Ongoing)

RC-PROC-002: Manual Data Entry Processes

Description: High reliance on manual data entry instead of automated data capture

Evidence:

- 45% of data in Salesforce CRM is manually entered (vs 15% industry benchmark for CRM systems)
- Legacy systems require 100% manual data entry

Impact Systems: Salesforce CRM, Archive System X, Old Billing System

Remediation:

- Implement web forms with automated validation (Target: Q1 2025)
- Deploy OCR technology for paper document digitization (Target: Q2 2025)
- Accelerate legacy system retirement (Target: Q4 2025)

RC-PROC-003: Inconsistent Data Governance

Description: Data quality standards and processes not consistently applied across all systems

Evidence:

- 58 systems, but only 12 have documented data quality SLAs
- Data stewards assigned to Tier 1 systems only (Tier 2 and Tier 3 lack dedicated stewardship)

Impact Systems: All Tier 2 and Tier 3 systems

Remediation:

- Expand data stewardship model to cover all 58 systems (Target: Q2 2025)
- Document data quality SLAs for all Tier 1 and Tier 2 systems (Target: Q1 2025)

6.2 Technology-Related Root Causes

RC-TECH-001: Lack of Preventive Validation Controls

Description: Applications lack real-time data validation at the point of entry

Evidence:

- 18 of 58 systems lack automated validation rules
- "Quick Lead" web form on public website has no email validation
- Date picker fields allow invalid default dates (e.g., 1900-01-01)

Impact Systems: Salesforce CRM, Patient Portal, Epic EHR, Legacy Systems

Remediation:

- Implement real-time validation for all web forms and application UI (Target: Q1-Q2 2025)
- Deploy validation APIs (email validation, phone number validation, address validation)

RC-TECH-002: Legacy System Technical Debt

Description: Legacy systems built 10-15 years ago lack modern data quality capabilities

Evidence:

- 3 legacy systems (Archive System X, Old Billing, Legacy Pharmacy) scheduled for retirement
- Legacy systems account for 38% of total data quality issues (despite representing only 12% of data volume)

Impact Systems: Archive System X, Old Billing System, Legacy Pharmacy

Remediation:

- Accelerate legacy system retirement (Target: Q2-Q4 2025)
- One-time data migration and cleansing projects

RC-TECH-003: Insufficient System Integration

Description: Systems operate in silos without real-time data synchronization

Evidence:

- Customer data exists in 3 systems (CRM, Portal, Billing) without synchronization
- Vendor data in Oracle EBS and SAP Ariba not synchronized (8.2% inconsistency rate)

Impact Systems: Cross-system consistency issues

Remediation:

- Implement Customer MDM (Target: Q3 2025)
- Continue Vendor MDM cleansing efforts (Target: Q1 2025)

6.3 People-Related Root Causes

RC-PPL-001: Insufficient Awareness of Data Quality Impact

Description: Staff underestimate the business impact of poor data quality

Evidence:

- Survey results: 42% of staff "strongly agree" that data quality is important (target: >80%)
- Anecdotal evidence: Staff often bypass validation rules to "save time"

Impact Systems: All systems with manual data entry

Remediation:

- Launch data quality awareness campaign (Target: Q1 2025)

- Publish case studies showing business impact of poor data quality
- Include data quality metrics in staff performance reviews

RC-PPL-002: Resource Constraints

Description: Data quality activities compete with operational priorities for limited staff time

Evidence:

- Data Quality team: 3 FTE (industry benchmark: 0.5% of employee count = 4.25 FTE for 850 employees)
- Data stewards spend <20% of time on data quality (target: 30-40%)

Impact: Reactive issue resolution instead of proactive quality management

Remediation:

- Increase Data Quality team staffing (Target: 5 FTE by Q2 2025)
- Formalize data steward time allocation (minimum 30% time on data governance/quality)

7. REMEDIATION PLAN

7.1 Critical Priority Issues (Q1 2025)

Issue ID	Issue Description	System	Owner	Target Date	Budget (SAR)	Status
DQ-FIX-001	Implement real-time email validation on "Contact Us" form	Sales CRM	IT Dev Lead	Jan 15, 2025	15,000	■ In Progress
DQ-FIX-002	Deploy email verification service to CRM date entry	Sales CRM	IT Dev Lead	Feb 1, 2025	30,000	■ Planned
DQ-FIX-003	Implement automated EHR verification at patient check-in	EMR/EHR	IT Ops Mgr	Feb 15, 2025	25,000	■ In Progress
DQ-FIX-004	Launch "Update Your Contact Info" patient portal	Patient Portal	Medical Services	Jan 30, 2025	10,000	■ Planned
DQ-FIX-005	Remove default date from EHR date picker	EMR/EHR	IT Dev Lead	Jan 20, 2025	8,000	■ In Progress
DQ-FIX-006	Implement date validation rules for DOB between 1900 and 2025	EMR/EHR	IT Dev Lead	Jan 20, 2025	5,000	■ In Progress
DQ-FIX-007	Execute one-time TRIM on multiple systems for 50,000 records	Multiple Systems	Database Admin	Mar 31, 2025	0	■ 30% Complete
DQ-FIX-008	Implement UI-level TRIM on multiple systems	Multiple Systems	IT Dev Lead	Mar 15, 2025	12,000	■ Planned
DQ-FIX-009	Vendor master data cleansing (100 duplicate vendors)	SAP	Procurement Mgr	Mar 31, 2025	0	■ 40% Complete
DQ-FIX-010	Document data quality SLAs for Tier 1 and Tier 2 issues	All Systems	DQ Manager	Mar 31, 2025	0	■ Planned

Q1 2025 Total Budget: SAR 105,000

7.2 High Priority Issues (Q2 2025)

Issue ID	Issue Description	System	Owner	Target Date	Budget (SAR)	Status
DQ-FIX-011	Implement automated data quality checks for Salesforce CRM	Salesforce CRM	CRM Lead	Apr 1, 2025	25,000	■ Planned
DQ-FIX-012	Conduct Salesforce duplicate record cleanup	Salesforce CRM	CRM Admin	June 30, 2025	80,000	■ Planned
DQ-FIX-013	Retrospective DOB collection for 15,000 call center contacts	Call Center	Call Center	June 30, 2025	45,000	■ Planned
DQ-FIX-014	Implement API integration for legacy systems to TDM	Legacy Systems	TDM Mgr	June 30, 2025	250,000	■ Pending Approval
DQ-FIX-015	Deploy OCR technology for paper document digitization	Archive System	Digitization Dir	June 30, 2025	180,000	■ Planned
DQ-FIX-016	Expand data stewardship to all 58 systems	All Systems	DSO	June 30, 2025	0	■ Planned

Q2 2025 Total Budget: SAR 580,000

7.3 Medium Priority Issues (Q3 2025)

Issue ID	Issue Description	System	Owner	Target Date	Budget (SAR)	Status
DQ-FIX-017	Implement Customer MDM for Systemica MDM	Systemica MDM	MDM Manager	Sep 30, 2025	450,000	■ Planned
DQ-FIX-018	Implement preventive DQ checks for legacy systems	Legacy Systems	Legacy Mgr	Sep 30, 2025	120,000	■ Planned
DQ-FIX-019	Accelerate legacy system migration to Archive System	Legacy System	System Modernization PM	Dec 31, 2025	1,200,000	■ Planned

Q3-Q4 2025 Total Budget: SAR 1,770,000

TOTAL REMEDIATION BUDGET (2025): SAR 2,455,000

7.4 Resource Requirements

Staffing Requirements

Team	Current FTE	Required FTE	Gap	Justification
Data Quality Team	3	5	+2	Industry benchmark: 0.5% of employee count
Data Stewards	5	15	+10	Expand coverage from Tier 1 to all 58 systems
Metadata Analysts	2	4	+2	Support metadata enrichment to 95% target
TOTAL	10	24	+14	

Budget Summary (2025)

Category	Amount (SAR)	% of Total
Technology (APIs, Tools)	580,000	24%

Professional Services (Consultants, Vendors)	730,000	30%
System Modernization (Legacy Retirement)	1,200,000	49%
TOTAL	2,455,000	100%

Budget Status: 65% approved (SAR 1,595,750); 35% pending approval (SAR 859,250)

8. DATA QUALITY TRENDS & ANALYTICS

8.1 Quarterly Trend Analysis (Q1-Q4 2024)

Quarter	Overall Score	Critical Issues	High Issues	Medium Issues	Low Issues	Total Issues
Q1 2024	85.2%	124	298	1,156	2,650	4,228
Q2 2024	87.5%	89	267	1,025	2,420	3,801
Q3 2024	89.0%	68	234	945	2,180	3,427
Q4 2024	92.4%	52	198	862	1,950	3,062
YoY Change (Q4 2023 - Q4 2024)	+10.2%	-67%	-42%	-28%	-33%	-33%

Trend Analysis: Consistent quarterly improvement in overall data quality score (+7.2% improvement from Q1 to Q4 2024), demonstrating effectiveness of systematic remediation efforts.

8.2 Issue Volume Trends

Issue Detection Method (Q4 2024)

Detection Method	Issue Count	% of Total
Automated Profiling (Informatica IDQ)	2,603	85%
User-Reported Issues	306	10%
Manual Data Audits	153	5%
TOTAL	3,062	100%

Key Insight: 85% of data quality issues detected proactively through automated profiling (up from 42% in Q1 2024), reducing reliance on reactive user-reported issues.

8.3 Resolution Performance

Issue Resolution Time (Q4 2024)

Issue Priority	SLA (Days)	Avg Resolution Time (Days)	SLA Compliance
Critical	3 days	2.1 days	96%
High	10 days	7.8 days	92%
Medium	30 days	24.5 days	88%
Low	90 days	78.2 days	85%

Overall Average Resolution Time: 8 days (down from 18 days in Q1 2024, a 56% improvement)

9. APPENDIX

9.1 Data Quality Rules Inventory

Company A maintains 1,240 active data quality rules across all systems, managed in Informatica Data Quality (IDQ) platform.

Rule Categories

Rule Category	Rule Count	Example Rule
Completeness Rules	342	"Patient National ID must not be NULL"
Accuracy Rules	285	"Mobile number must pass SMS verification test"
Consistency Rules	198	"Vendor Name in Oracle EBS must match Vendor Name
Validity Rules	245	"Email address must match regex: ^{a-zA-Z0-9._%+-}
Uniqueness Rules	125	"Invoice Number must be unique in Financial System
Timeliness Rules	45	"Patient contact info must be updated within 12 mo
TOTAL	1,240	

Sample Data Quality Rules

- ##### Rule ID: DQ-R-001 - Patient National ID Completeness
- Dimension: Completeness

- System: Epic EHR

- Field: Patient.NationalID

- Rule Logic: `Patient.NationalID IS NOT NULL AND Patient.NationalID != ``

- Severity: Critical (Priority 1)
- SLA: 100% compliance
- Current Compliance: 98.0%

Rule ID: DQ-R-018 - Email Address Validity

- Dimension: Validity
- System: Salesforce CRM
- Field: Customer.Email
- Rule Logic: `REGEX\_MATCH(Customer.Email, '^[a-zA-Z0-9.\_%+~]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}\$')`
- Severity: High (Priority 2)
- SLA: 95% compliance
- Current Compliance: 74.8% (BELOW TARGET - remediation required)

Rule ID: DQ-R-042 - Vendor Name Consistency

- Dimension: Consistency
- System: Oracle EBS + SAP Ariba
- Field: Vendor.VendorName
- Rule Logic: `Oracle\_EBS.Vendor.VendorName = SAP\_Ariba.Vendor.VendorName WHERE VendorID matches`
- Severity: Medium (Priority 2)
- SLA: 95% consistency
- Current Compliance: 91.8%

Rule ID: DQ-R-089 - Invoice Number Uniqueness

- Dimension: Uniqueness
- System: Oracle EBS
- Field: Invoices.InvoiceNumber
- Rule Logic: `COUNT(DISTINCT InvoiceNumber) = COUNT(InvoiceNumber)`
- Severity: Critical (Priority 1)
- SLA: 100% uniqueness
- Current Compliance: 100% ■

9.2 Profiling Statistics Summary

Overall Profiling Statistics (Q4 2024)

Metric	Value
Total Records Profiled	150,000,000
Total Tables/Files Profiled	898
Total Data Volume	2,367 GB
Profiling Duration	8 weeks
Automated Profiling Jobs Executed	1,840 jobs

Manual Validation Samples	28,400 records
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System-Level Profiling Details

System	Records Profiled	Tables	Duration	DQ Score
Oracle Data Warehouse	142,000,000	380	6 weeks	95.4%
Cerner HIS	5,200,000	95	2 weeks	94.8%
Oracle EBS	2,800,000	68	2 weeks	99.1%
Epic EHR	18,500	142	1 week	96.2%
Salesforce CRM	95,000	48	1 week	79.5%
All Other Systems	1,286,500	165	4 weeks	Various

9.3 Glossary of Terms

Term	Definition
Completeness	The extent to which required data is present (no m
Accuracy	The extent to which data correctly represents the
Consistency	The extent to which data is consistent across syst
Validity	The extent to which data conforms to defined forma
Uniqueness	The extent to which records are unique (no duplica
Timeliness	The extent to which data is current and up-to-date
Critical Data Element (CDE)	Data element with high business criticality requir
Data Quality Rule	Automated check to validate data against defined q
Data Profiling	Statistical analysis of data to assess quality and
Root Cause Analysis (RCA)	Systematic investigation to identify underlying ca
CAPA	Corrective and Preventive Action - structured reme
SLA	Service Level Agreement - target performance metri

END OF REPORT

Company A - Internal

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